Week 10 Lecture 3 STAT2004/2904/7004

- Ass 3 due today @ 4pm
- Ass 4 release on Mon 12 Oct

Recap:

(1) For testing $H_0: p = p^*$ vs. $H_1: p \neq p^*$ for categorical data, the generalized likelihood ratio test (GLRT) is equivalent to large values of

$$-2 \log \Lambda = 2 \sum_{\text{cells } k} O_k \log\left(\frac{O_k}{E_k}\right)$$

(2) The GLRT is asymptotically equivalent to the classical Pearson's $\chi^2$ statistic:

$$\chi^2 = \sum_{\text{cells } k} \frac{(O_k - E_k)^2}{E_k}$$

(3) Fisher: "[A null hypothesis] is never proved or established, but it is possibly disproved, in the course of experimentation."

Example 8.4 (Mendel's genetics data, continued)

| Type | Mendel's theory | Mendel's counts | Alan's count |
|---|---|---|---|
| smooth yellow | 9/16 | 315 | 901 |
| smooth green | 3/16 | 108 | 298 |
| wrinkly yellow | 3/16 | 102 | 302 |
| wrinkly green | 1/16 | 31 | 99 |
| | | | 1600 |

The exact GLRT here is $-2 \log \Lambda = 2 \sum_{\text{cells } k} O_k \log\left(\frac{O_k}{E_k}\right) = 0.618$

& Pearson's $\chi^2$ here is $\chi^2 = \sum_{\text{cells } k} \frac{(O_k - E_k)^2}{E_k} = 0.604$

Both of these have an approximate $\chi^2_3$ distn, & so the cutoff we need at the 5% level to reject the would be:

```
> qchisq(0.95, 3)
[1] 7.814
```

Hence, we decide to accept Mendel's theory (p-value = 0.9)

Our formal analysis of Mendel's data would stop here (in 1st year). However, Fisher noticed that a p-value of 0.90 was rather large. This is because Mendel's counts were very well aligned with his theoretical proportions. This motivated Fisher to examine ALL of Mendel's log books. Pooled together all counts from all of Mendel's experiments, Fisher found a p-value of 0.9996 for testing Mendel's theory. What kind of counts did Mendel have? A little bit like Alan's counts.

While getting a small p-value makes us suspicious about the null hypothesis, getting a very large p-value also makes us suspicious ... about either data-fabrication and/or pseudo-replication.

Data fabrication--

Suppose we are sampling from a pop^n with an underlying proportion of 9/16 for some trait

Real data might look like:

$\frac{12}{16} \longrightarrow \frac{82}{160} \longrightarrow \frac{847}{1600}$ ← actual counts diverge from 900/1600

but proportion → 9/16

Fake data

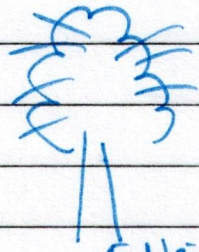
$\frac{12}{16} \longrightarrow \frac{88}{160} \longrightarrow \frac{901}{1600}$

not enough variability in the data!

## Pseudo-replication

Consider comparing two fertilizers using 2 trees:

Experiment ①




fertilizer 1



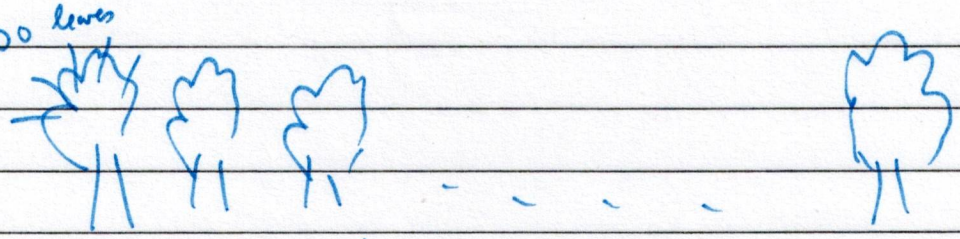

fertilizer 2

" Claim: my experiment had N=2000 samples!

This would be an example of Pseudo-replication: multiple measurements are taken on each experimental unit but we pretend they were individual measurements from independent units.

Experiment ② N=2000 trees.

n=100 leaves



fertilizer 1 fertilizer 2

This would give us 2000 genuine replications.

Of course, experiment ② is much more expensive than experiment ①.

## Working with two (or more) categorical factors.

Pearson's $\chi^2$ (or GLRT) test of independence & test of homogeneity are carried out in exactly the same way, but the underlying hypotheses are slightly different & the way the data are collected is slightly different

In a test of independence (STAT1201/1301), units are sampled from a single pop & then cross-classified according to two (or more) factors. The null hypothesis is that these two factors are independent / unrelated:

$$H_0: p_{ij} = p_{i\cdot} \times p_{\cdot j} \quad \forall\, i,j$$

$p_{ij}$: proportion in level $i$ of factor 1 & level $j$ of factor 2

where

$p_{i\cdot}$ = overall proportion in level $i$ of factor 1

$p_{\cdot j}$ = overall proportion in level $j$ of factor 2.

versus $H_1: p_{ij} \neq p_{i\cdot} \times p_{\cdot j}$ for at least one pair of $i$ & $j$

We have done this a lot in first year already (so I will give a tutorial question to refresh this), hence I want focus more on test of homogeneity

In a test of homogeneity, instead of sampling from a single popn & cross-classifying, we sample separately from two (or more) subpopn's & we compare occurrence of a certain categorical factor between these subpopn's. We are interested in testing whether the distn of this categorical factor is homogeneous (ie. same) across our (sub)populations

For example, when comparing prevalence of a certain genetic / biological trait across species (e.g. humans vs mice), we sample separately from our (sub)populations & then we compare their traits.

Here is a more subtle example:

Example 8.6. All UQ academic staff are provided a computer, & staff can choose between windows (W), Mac (M) & Linux (L). To plan for next year's IT intake, the choices of all* staff (across 3 biggest faculties Science, Engineering, Arts) who were provided a new computer last year was collected & tabulated below:

| | Faculty | | | | | | |
|---|---|---|---|---|---|---|---|
| OS | Science | | Eng. | | Arts | | Total |
| Windows | 12 | (12.42) | 17 | (13.8) | 9 | | 38 |
| Mac | 8 | (10.53) | 9 | (11.7) | 15 | | 32 |
| Linux | 7 | | 4 | | 1 | | 12 |
| Total | 27 | | 30 | | 25 | | 82 |

Q: Is OS preference homogeneous (same) across the three faculties?

Null hypothesis here is:

$$H_0: \begin{pmatrix} p_W \\ p_M \\ p_L \end{pmatrix}_{\text{Science}} = \begin{pmatrix} p_W \\ p_M \\ p_L \end{pmatrix}_{\text{Eng}} = \begin{pmatrix} p_W \\ p_M \\ p_L \end{pmatrix}_{\text{Arts}} = \begin{pmatrix} p_W \\ p_M \\ p_L \end{pmatrix}_{\text{Common}}$$

$$\vec{p}_{\text{Science}} = \vec{p}_{\text{Eng}} = \vec{p}_{\text{Arts}} = \vec{p}_{\text{common}}$$

versus alternative hypothesis:

$H_1$: at least probability vector is different
OR not all probability vectors are the same.
OR there is NO restrictions on each of the prob. vectors

How many free parameters under the alternative model:
$6 = 3 \times 2$
(in general, $c(r-1)$, where $c = \#$ columns, $r = \#$ rows)

How many free parameters under the null model? 2
(in general, $(r-1)$)

How many degrees-of-freedom would a test of homogeneity take up?

$df: 6 - 2 = 4 = c(r-1) - (r-1)$
$= (c-1)(r-1)$

Under the null, what would be the best guess (MLE) of the common prob. vector:

$$\begin{pmatrix} \hat{p}_W \\ \hat{p}_M \\ \hat{p}_L \end{pmatrix}_{\text{common}} = \begin{pmatrix} 38/82 \\ 32/82 \\ 12/82 \end{pmatrix} = \begin{pmatrix} 0.46 \\ 0.39 \\ 0.15 \end{pmatrix}$$

So out of 27 service staff, how many would you expect to prefer windows?

$$27 \times 0.46 = 12.42$$

" " " " " " "

" Mac?

$$27 \times 0.39 = 10.53$$

So out of 30 Engineering staff, how many " "

" windows?

$$30 \times 0.46 = 13.8$$

" " " " " " "

" Mac?

$$30 \times 0.39 = 11.7$$

We can write these expected counts (in red) next to our observed counts in the table.